Quantitative Image-Processing Analysis of High-Speed Droplet Impact and Bounce Dynamics on a Superhydrophobic Surface

A MATLAB Image-Processing Based Project Report

Submitted by

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1. Methodology

1.1 Image-Processing Pipeline Workflow

The developed MATLAB pipeline utilizes a sequential logic to transform raw optical data into quantitative measurements. The workflow consists of five main stages, described below.

1. Preprocessing: Raw frames are converted to grayscale. Background subtraction is executed via imabsdiff using a reference frame (Frame 30) to isolate the moving droplet from the static experimental apparatus.
2. Segmentation: A binary mask is generated through thresholding (pixel < threshold). This isolates the dark droplet mass as logical 1 (white) against a logical 0 (black) background.
3. Morphological Cleanup: imclose uses a disk-structuring element (radius 25) to bridge gaps in the droplet boundary caused by motion blur or thin fluid walls; imfill treats internal “holes” caused by lighting reflections as part of the fluid mass, creating a solid object; and bwareafilt ensures only the largest object (the droplet) is analyzed, filtering out secondary satellites or sensor noise.
4. Boundary Detection: The bwboundaries function extracts the continuous droplet perimeter for geometric tracking.

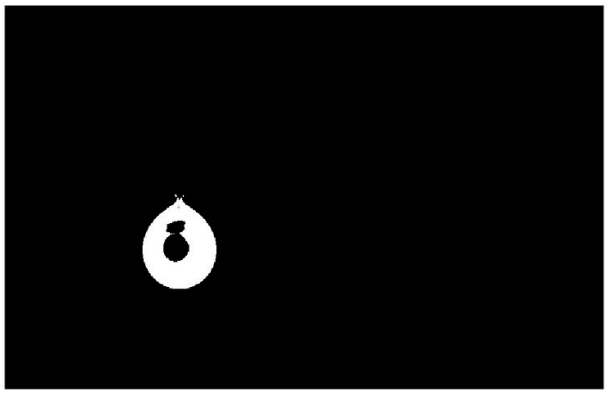


Image 1: Representative raw / preprocessed frame

1.2 Key Computational and Parameter Choices

To ensure robust and continuous tracking of the droplet across all physical phases, several critical parameter-driven decisions were made.

First, Frame 30 was selected as the static reference background. This specific frame was chosen because the three core objects of the experiment — the water droplet, the dispensing dropper (nozzle), and the superhydrophobic substrate — are completely separated and visually distinct in the spatial coordinate system. This distinct separation made it easy to perform absolute background subtraction and isolate the moving droplet without boundary overlap.

Second, the binarization threshold was optimized by evaluating a wide spectrum of values ranging from 10 to 50 across different video phases. Testing revealed a clear physical trade-off: lower thresholds, such as 10, yielded poor droplet boundary definition during the initial high-contrast stages of free-fall compared to a sharper threshold of 50. However, during the extreme impact and rebound phases, the droplet's fluid walls stretch and thin out, becoming semi-transparent; under these conditions, a lower threshold of 10 was strictly required to accurately track the fading droplet boundary without losing critical tracking data.

Finally, to address the severe boundary breaks caused by internal camera light reflections, morphological closing was applied using a circular structuring element. Structuring element disk radii were iteratively tested between 10 and 25. The evaluation demonstrated that utilizing a higher disk radius (up to 25) was essential, as the larger circular "stamp" forcefully bridged and fused the broken boundaries of the deformed droplet. This high disk radius successfully closed the fragmented mask edges, enabling the subsequent hole-filling operations to restore a solid, mathematically continuous droplet silhouette for precise spatial measurement.

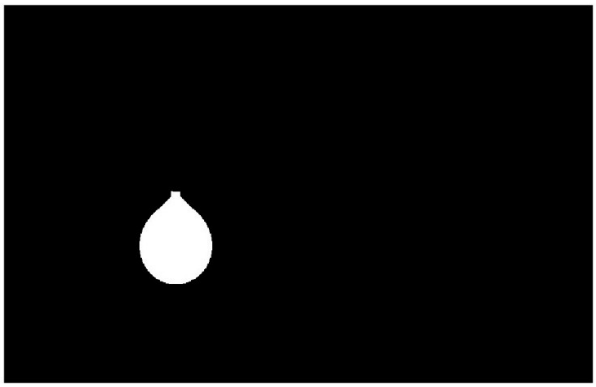


Image 2: Segmentation / morphological cleanup stage

1.3 Spatial Image Calibration

The high-speed video has no scale reference, and the red scale bar in the guideline photograph was taken at a different magnification. Calibration was therefore done directly from the video, using the substrate as a reference of known size. In Frame 1, only the bottom third was analyzed, to leave out the nozzle and droplet. Otsu's method (graythresh) thresholded the region, the mask was inverted, and bwareafilt kept only the substrate. regionprops measured the substrate's width in pixels, and dividing its known visible width (15 mm) by that count gave a calibration factor of 15 mm/pixel. This factor converted all area, diameter and centroid measurements to millimeters.

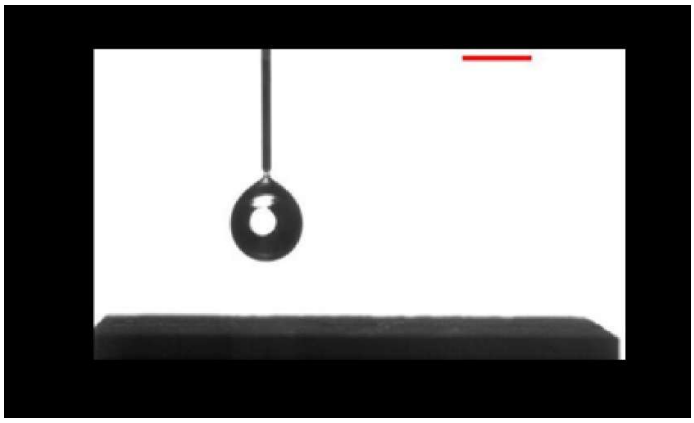


Image 3: Spatial calibration reference frame

2. Results

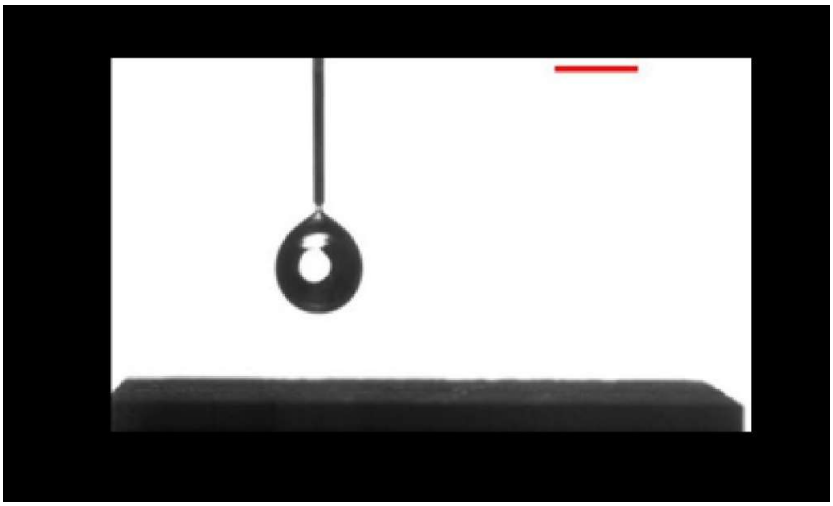


Figure1

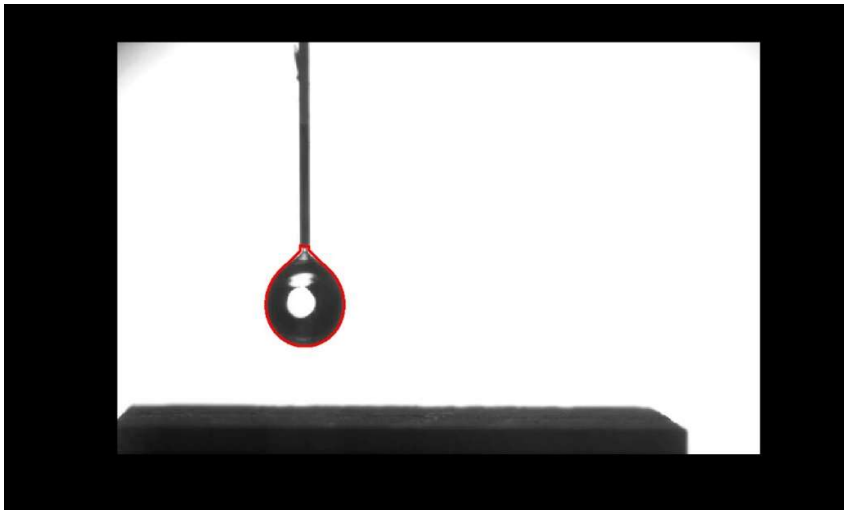


Figure2

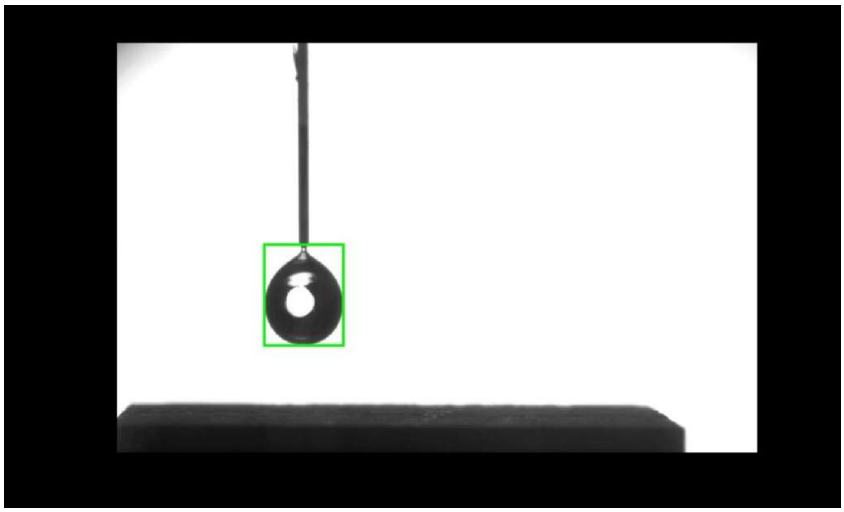


Figure3

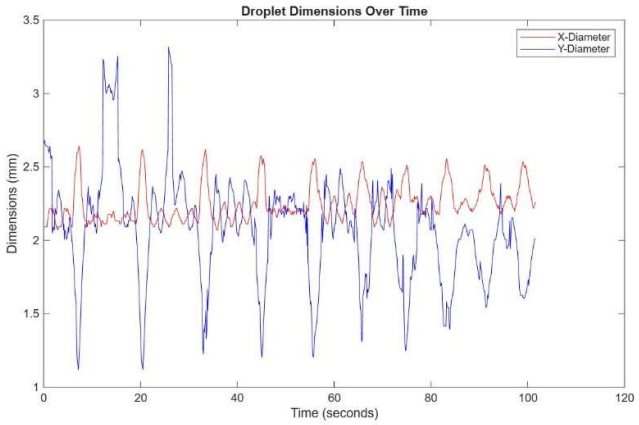


Figure4

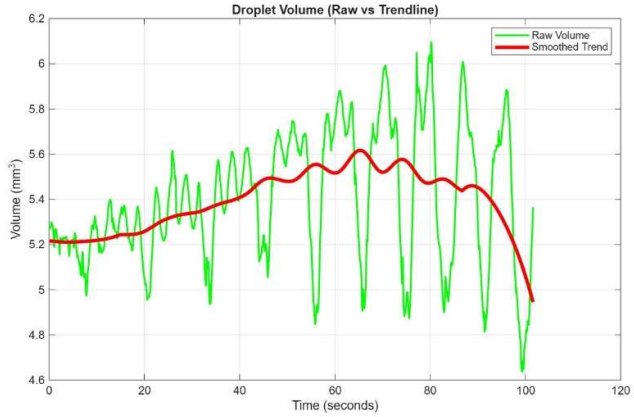


Figure5

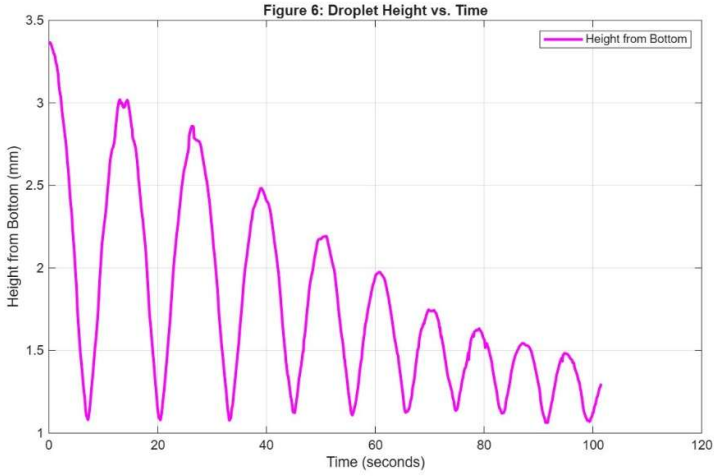


Figure6

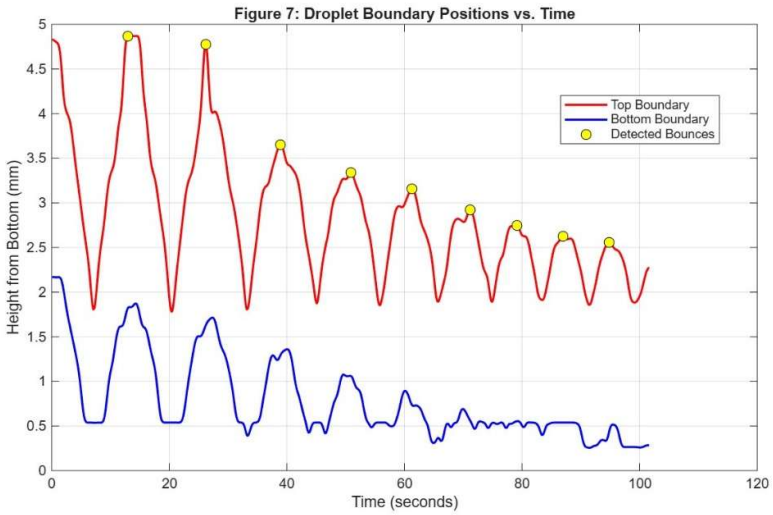
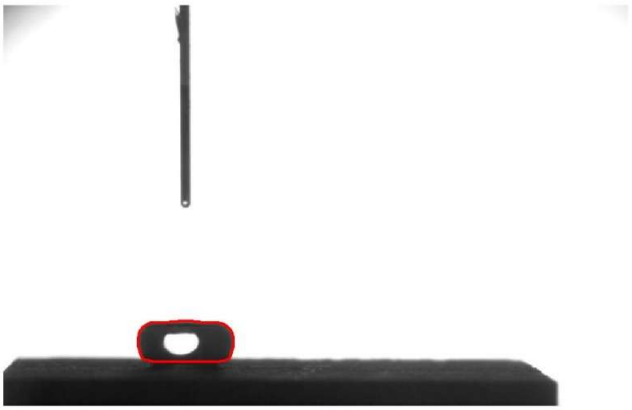
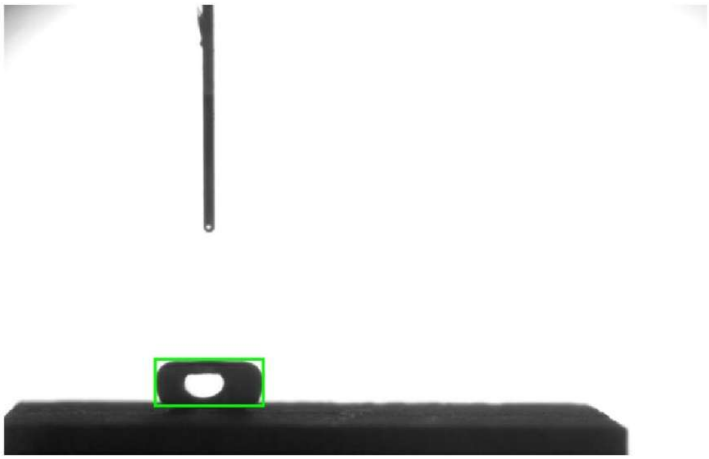


Figure7



Boundary Detection on Frome-72



Bounding Box Detection on Frome-72

3. Analysis & Interpretation

3.1 Droplet Deformation and Geometric Evolution

The physical deformation of the water droplet throughout the high-speed experiment is quantitatively captured by the evolution of its horizontal (x) and vertical (y) diameters (**Figure 4**). During the initial free-fall phase, both diameters remain nearly identical, confirming that the droplet maintains a highly spherical shape dictated by surface tension minimization.

The most dramatic geometric divergence occurs during the initial impact phase and the subsequent rebound events. Upon first contact with the superhydrophobic substrate, the vertical diameter (y) plunges rapidly toward a minimum value while the horizontal diameter (x) surges to a maximum. This indicates severe flattening into an oblate spheroid (or flat disc-like geometry), driven by the conversion of axial kinetic energy into radial fluid momentum. Conversely, during the rebound detachment phase, the droplet undergoes extreme vertical stretching, where the vertical diameter significantly exceeds the horizontal diameter, transforming into a highly elongated prolate shape.

These cyclic transitions represent a highly dynamic exchange of kinetic energy and surface elastic energy. The droplet recovers its nearly spherical shape only during the peak apex of the higher rebounds, where the vertical and horizontal dimensions briefly converge. As the bounces decay, the droplet transitions into a resting settling phase. During this phase, high-frequency, low-amplitude capillary wave oscillations are observed until viscous forces completely damp the remaining surface energy and restore the droplet to a stable, static spherical dome.

3.2 Droplet Volume Conservation and 2D Projection Limitations

Because water is an incompressible fluid, the droplet's true physical volume must remain constant throughout the experiment. However, a single camera records only a 2D silhouette, so any 3D volume must be reconstructed from an assumption about the hidden depth (z-axis). How good that assumption is determines how much the calculated volume oscillates as the droplet deforms.

Three reconstruction methods were evaluated:

- The Spheroid Formula Fallacy ($V = \pi/6 \cdot x^2y$): This formula squares the horizontal diameter, so it is very sensitive to it. When the droplet flattens radially on impact, x grows, and because the formula assumes the hidden depth grows to match x, the calculated volume spikes unrealistically. When the droplet stretches into a prolate shape, x shrinks and the calculated volume drops sharply.

- The Area-Equivalent Sphere Method ($r = \sqrt{(A/\pi)}$, $V = 4/3 \cdot \pi r^3$): Counting the white pixels in the binary mask gives the true projected area, which avoids the empty corners of a bounding box. However, it still forces every frame into a sphere, regardless of the droplet's actual shape. During impact flattening this badly underestimates the volume: it varied between 3.209 and 6.314 µL (mean 5.080 µL), a swing of ±30.6% about the mean, and dipped 44.9% below the initial free-fall volume.

- The Pappus–Guldinus Method (adopted): Instead of assuming a sphere, this method uses the droplet's actual measured profile. The droplet is treated as axisymmetric about a vertical axis through its centroid, which is reasonable for an impact normal to a flat substrate. By the Pappus–Guldinus theorem, revolving one half of the silhouette about this axis gives the volume: $V = 2\pi \cdot \bar{r} \cdot A_half$ where A_half is the area of the half-profile (counted in pixels) and $\bar{r}$ is the horizontal distance of that half's centroid from the axis. The volume was computed separately for the left and right halves and averaged, to reduce the effect of small left–right asymmetry.

Because the method follows the real boundary, it handles oblate and prolate shapes without a fixed shape factor.

Result. The Pappus–Guldinus method kept the calculated volume between 4.636 and 6.095 µL (mean 5.394 µL), a swing of ±13.5% about the mean, compared with ±30.6% for the area-equivalent sphere. That is a 56% reduction in volume-estimation error (total swing reduced from 61.1% to 27.0% of the mean), achieved with a single camera.

Remaining error. The leftover ±13.5% oscillation follows the impact and rebound cycles. It comes from two sources the method cannot remove: (1) during violent deformation the droplet is not perfectly axisymmetric, so revolving a 2D profile cannot capture its true 3D shape; and (2) out-of-plane drift toward the camera defocuses the edges and degrades the binary mask. A 300-frame local regression ('loess') trendline is overlaid on Figure 5 to show the mean volume level. It is a visual guide to the average, not a correction. Removing the remaining error fully would need the orthogonal dual-camera setup proposed in Section 4.2.

3.3 Kinematics, Bounce Behaviour, and Energy Conversion

The vertical centroid trajectory plotted in **Figure 6** and the top/bottom boundary tracking in **Figure 7** provide a comprehensive map of the energy exchanges occurring during impact.

During free-fall, the vertical centroid (y_centroid) follows a smooth, parabolic downward trajectory, demonstrating constant gravitational acceleration. Upon contact, the bottom boundary (y_bottom) is instantly arrested and pinned to the solid substrate boundary condition (flattening out at the substrate level, as seen in **Figure 7**). However, the top boundary (y_top) continues to move downward rapidly, carrying the inertia of the upper fluid mass. This compression phase converts kinetic energy into internal fluid strain energy (surface energy) as the droplet flattens and its surface area dramatically increases. The stored surface energy then acts as a restoring force, pulling the droplet back upward and forcing the bottom boundary (y_bottom) to lift off the surface to initiate a rebound.

The vertical centroid trajectory and the detected bounce peaks show a progressive decay in rebound height after each of the 9 observed bounces. This decay is driven by several irreversible thermodynamic and mechanical loss mechanisms:

- Viscous Dissipation: Internal shear stresses and friction within the water molecule layers dissipate kinetic energy into low-grade thermal energy as the droplet deforms and wobbles.
- Contact-Line Friction and Hysteresis: Energy is lost at the solid-liquid interface due to molecular pinning and three-phase contact line friction during spreading and retraction.

Despite these continuous losses, the physical observation of 9 distinct bounces highlights the extreme effectiveness of the superhydrophobic aluminum surface. If the surface were hydrophilic, the high surface energy of the solid would cause the droplet to instantly spread and wet the substrate, completely suppressing any rebound behaviour. The high density of successful rebounds quantitatively validates the exceptionally low roll-off angle and high contact angle achieved by the eco-friendly immersion coating.

4. Method Evaluation

4.1 Experimental Difficulties and Sources of Error

- Optical Boundary Breakdown via Out-of-Plane Motion: As the droplet bounced, it drifted slightly forward in 3D space, moving closer to the camera lens and out of the optimal focal plane. This forward movement altered the backlighting path, causing the droplet's outer surface and boundaries to appear extremely faint and low-contrast. This physical defocusing led to a severe breakdown in the binary mask, as standard thresholding could no longer resolve the faint edges.
- 2D Projection and Volumetric Limitations: Single-camera 2D optical metrology cannot capture out-of-plane 3D fluid displacement (along the hidden z-axis). Consequently, estimating 3D volume using 2D dimensions during extreme oblate/prolate transitions introduced severe geometric calculation errors (the "spheroid fallacy").
- Contact-Line and Substrate Noise: Intense light reflections off the aluminum substrate created 1-pixel jagged fluctuations (optical noise) along the droplet's bottom boundary when resting on the surface. To prevent these micro-vibrations from triggering false bounce detections in the peak-finding algorithm, a 15-frame Gaussian smoothing filter was required, and bounce tracking was shifted entirely to the cleaner top boundary.

4.2 Proposed Future Improvements

1. Dynamic ROI and Adaptive Thresholding: Implementing a dynamic Region of Interest (ROI) tracking window paired with Otsu's adaptive thresholding (graythresh) would allow the threshold to adjust automatically to local lighting variations, eliminating manual threshold switching.
2. Dual-Camera 3D Metrology: Utilizing a synchronized, orthogonal dual-camera setup would allow full 3D reconstruction of the droplet's surface. This would completely eliminate 2D projection errors and provide exact, direct volume measurements during violent deformations.
3. Active Machine Learning Segmentation: Replacing morphological closing with a lightweight convolutional neural network (CNN) or K-Means clustering for pixel classification would drastically improve boundary extraction on faint, semi-transparent liquid walls without distorting the physical geometry.

5. Conclusions

- Bounces: 9 distinct bounces recorded.
- Energy Loss: 17.37% of the droplet's mechanical energy was dissipated on the first impact. The centroid rose to 1.715 mm above its resting level on the first rebound, from an initial drop height of 2.075 mm, giving a coefficient of restitution of about 0.91.
- Deformation Limits: At maximum compression during the first collision, the droplet reached a peak horizontal width (X-diameter) of 2.641 mm and a minimum vertical height (Y-diameter) of 1.120 mm. That is a spreading ratio of about 1.21 and a compression to about 51% of the equivalent spherical diameter (2.18 mm).
- Centroid Stabilization: By the end of the video, the droplet's centre of mass settled at a resting height of 1.012 mm above the impact surface.
- Volume Tracking (Pappus–Guldinus): The reconstructed volume averaged 5.407 mm^3 during the active bouncing phase and 5.220 mm^3 after settling. The two differ by only 3.5%, consistent with conservation of the incompressible droplet's volume within the limits of single-camera measurement.

6. Declaration

I declare that I take full responsibility for the analysis, results, MATLAB code, interpretations, and conclusions presented in this report and the accompanying presentation.

I may have consulted various resources during the development of this project. However, I have reviewed and understood the material used in my work, adapted it to my requirements, and confirm that I am able to explain and defend the methodology, code, results, and conclusions presented by me. I further confirm that the submitted work represents my own independent analysis and has not been copied from the work, code, or results of another student.

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Signature:

Date: 31-08-2026

References

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